

Java Practice Workbook (Problems 1-26)

1. Check Whether Age is Greater Than 60

Problem Statement: A government office wants to identify senior citizens eligible for special benefits. Read a person's age and determine whether the person is a senior citizen. If the age is greater than 60, display 'Senior Citizen'; otherwise display 'Not a Senior Citizen'.

Input: Age (integer)

Output: Senior Citizen

Sample Input: 65

Sample Output: Senior Citizen

Conditions / Constraints: Age is positive. Use only if.

Hints: Read age, compare with 60.

2. Student Pass/Fail

Problem Statement: Write a Java program to solve the problem: Student Pass/Fail. Design the program using the concepts taught in class and display meaningful output.

Input: Marks

Output: Pass

Sample Input: 72

Sample Output: Pass

Conditions / Constraints: Marks 0-100. Use if-else.

Hints: Compare with 35.

3. Bank Loan Eligibility

Problem Statement: Write a Java program to solve the problem: Bank Loan Eligibility. Design the program using the concepts taught in class and display meaningful output.

Input: Age, Salary

Output: Loan Approved

Sample Input: 25, 35000

Sample Output: Loan Approved

Conditions / Constraints: Age $\geq$ 21 and Salary $\geq$ 30000.

Hints: Use &&.

4. User Login

Problem Statement: Write a Java program to solve the problem: User Login. Design the program using the concepts taught in class and display meaningful output.

Input: Username, Password

Output: Login Successful

Sample Input: admin, java123

Sample Output: Login Successful

Conditions / Constraints: Use .equals().

Hints: Validate both.

5. Income Tax Slab

Problem Statement: Write a Java program to solve the problem: Income Tax Slab. Design the program using the concepts taught in class and display meaningful output.

Input: Annual Income

Output: 20% Tax

Sample Input: 700000

Sample Output: 20% Tax

Conditions / Constraints: Use if-else-if.

Hints: Check slabs.

6. Vowel or Consonant

Problem Statement: Write a Java program to solve the problem: Vowel or Consonant. Design the program using the concepts taught in class and display meaningful output.

Input: Character

Output: Vowel

Sample Input: a

Sample Output: Vowel

Conditions / Constraints: Accept upper/lower.

Hints: Compare vowels.

7. Print ASCII Values of A-Z

Problem Statement: Write a Java program to solve the problem: Print ASCII Values of A-Z. Design the program using the concepts taught in class and display meaningful output.

Input: -

Output: A 65 ... Z 90

Sample Input: -

Sample Output: A 65 ... Z 90

Conditions / Constraints: Use loop.

Hints: Loop A to Z.

8. Countdown

Problem Statement: Write a Java program to solve the problem: Countdown. Design the program using the concepts taught in class and display meaningful output.

Input: Starting Number

Output: 5 4 3 2 1

Sample Input: 5

Sample Output: 5 4 3 2 1

Conditions / Constraints: Use loop.

Hints: Decrement.

9. Days in Each Month

Problem Statement: Write a Java program to solve the problem: Days in Each Month. Design the program using the concepts taught in class and display meaningful output.

Input: -

Output: Print all valid days

Sample Input: -

Sample Output: Print all valid days

Conditions / Constraints: Nested loops.

Hints: Outer month, inner day.

10. Search First Match

Problem Statement: Write a Java program to solve the problem: Search First Match. Design the program using the concepts taught in class and display meaningful output.

Input: Array, Target

Output: Found at index 2

Sample Input: 10 20 30 / 30

Sample Output: Found at index 2

Conditions / Constraints: Use break.

Hints: Stop at first match.

11. Power Using Method

Problem Statement: Write a Java program to solve the problem: Power Using Method. Design the program using the concepts taught in class and display meaningful output.

Input: Base, Exponent

Output: 32

Sample Input: 2, 5

Sample Output: 32

Conditions / Constraints: Don't use Math.pow().

Hints: Repeated multiplication.

12. cube() Calls square()

Problem Statement: Write a Java program to solve the problem: cube() Calls square(). Design the program using the concepts taught in class and display meaningful output.

Input: Number

Output: 64

Sample Input: 4

Sample Output: 64

Conditions / Constraints: Reuse square().

Hints: cube=square*n.

13. average() Calls sum()

Problem Statement: Write a Java program to solve the problem: average() Calls sum(). Design the program using the concepts taught in class and display meaningful output.

Input: Three Numbers

Output: 20

Sample Input: 10 20 30

Sample Output: 20

Conditions / Constraints: Reuse sum().

Hints: Average=sum/3.

14. areaOfSquare() Calls square()

Problem Statement: Write a Java program to solve the problem: areaOfSquare() Calls square(). Design the program using the concepts taught in class and display meaningful output.

Input: Side

Output: 36

Sample Input: 6

Sample Output: 36

Conditions / Constraints: Reuse square().

Hints: Area=side².

15. percentage() Calls totalMarks()

Problem Statement: Write a Java program to solve the problem: percentage() Calls totalMarks(). Design the program using the concepts taught in class and display meaningful output.

Input: Five Marks

Output: 80%

Sample Input: 80 70 90 60 100

Sample Output: 80%

Conditions / Constraints: Out of 500.

Hints: Reuse totalMarks().

16. largest() Calls max() Twice

Problem Statement: Write a Java program to solve the problem: largest() Calls max() Twice. Design the program using the concepts taught in class and display meaningful output.

Input: Three Numbers

Output: 40

Sample Input: 25 40 35

Sample Output: 40

Conditions / Constraints: Call max() twice.

Hints: Compare result with third.

17. Student Result System

Problem Statement: Write a Java program to solve the problem: Student Result System. Design the program using the concepts taught in class and display meaningful output.

Input: Student details

Output: Result Card

Sample Input: -

Sample Output: Result Card

Conditions / Constraints: Compute total, %, grade.

Hints: Use methods.

18. ATM Menu

Problem Statement: Write a Java program to solve the problem: ATM Menu. Design the program using the concepts taught in class and display meaningful output.

Input: Menu Choice

Output: ATM Operations

Sample Input: -

Sample Output: ATM Operations

Conditions / Constraints: Use switch.

Hints: Repeat until Exit.

19. Banking Menu

Problem Statement: Write a Java program to solve the problem: Banking Menu. Design the program using the concepts taught in class and display meaningful output.

Input: Menu Choice

Output: Banking Operations

Sample Input: -

Sample Output: Banking Operations

Conditions / Constraints: Switch + methods.

Hints: Separate methods.

20. Library Book Search

Problem Statement: Write a Java program to solve the problem: Library Book Search. Design the program using the concepts taught in class and display meaningful output.

Input: Book Name

Output: Book Found

Sample Input: Java

Sample Output: Book Found

Conditions / Constraints: Use array.

Hints: equalsIgnoreCase().

21. Employee Salary Calculator

Problem Statement: Write a Java program to solve the problem: Employee Salary Calculator. Design the program using the concepts taught in class and display meaningful output.

Input: Basic,HRA,DA

Output: Gross Salary

Sample Input: 10000 2000 1500

Sample Output: Gross Salary

Conditions / Constraints: Gross=Basic+HRA+DA.

Hints: Method.

22. Electricity Bill Generator

Problem Statement: Write a Java program to solve the problem: Electricity Bill Generator. Design the program using the concepts taught in class and display meaningful output.

Input: Units

Output: Bill Amount

Sample Input: 250

Sample Output: Bill Amount

Conditions / Constraints: Use slabs.

Hints: if ladder.

23. Marks Card Generator

Problem Statement: Write a Java program to solve the problem: Marks Card Generator. Design the program using the concepts taught in class and display meaningful output.

Input: Student details

Output: Marks Card

Sample Input: -

Sample Output: Marks Card

Conditions / Constraints: Display total,%,grade.

Hints: Reuse methods.

24. Mini Calculator

Problem Statement: Write a Java program to solve the problem: Mini Calculator. Design the program using the concepts taught in class and display meaningful output.

Input: Choice and Numbers

Output: Result

Sample Input: -

Sample Output: Result

Conditions / Constraints: Switch + methods.

Hints: Separate methods.

25. User Login System

Problem Statement: Write a Java program to solve the problem: User Login System. Design the program using the concepts taught in class and display meaningful output.

Input: Username, Password

Output: Success/Locked

Sample Input: admin java123

Sample Output: Success/Locked

Conditions / Constraints: 3 attempts.

Hints: Loop.

26. Menu-driven Array Operations

Problem Statement: Write a Java program to solve the problem: Menu-driven Array Operations. Design the program using the concepts taught in class and display meaningful output.

Input: Array and Menu

Output: Menu Output

Sample Input: -

Sample Output: Menu Output

Conditions / Constraints: Switch + methods.

Hints: Read array once.